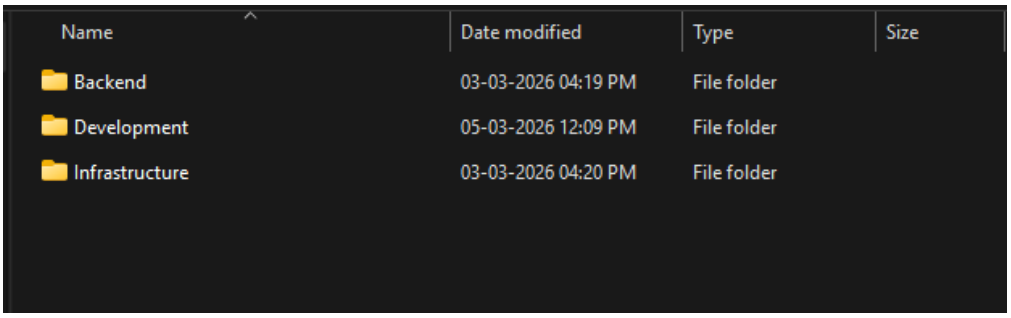
Terraform Task - 06

- Modules:
- 

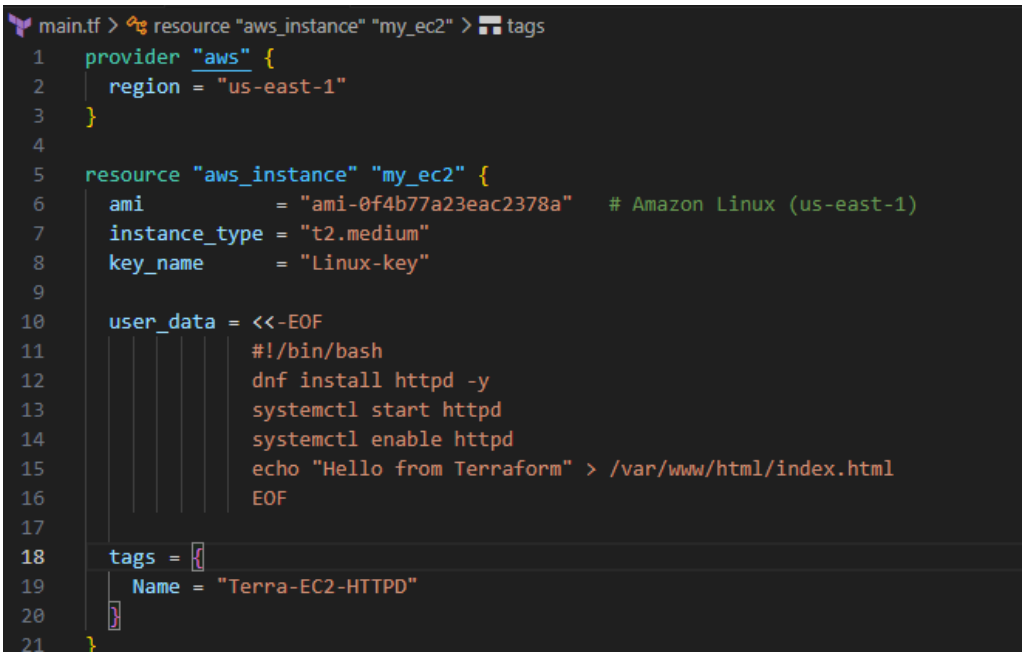
Name	Date modified	Type	Size
Backend	03-03-2026 04:19 PM	File folder	
Development	05-03-2026 12:09 PM	File folder	
Infrastructure	03-03-2026 04:20 PM	File folder	

- Backend is configured in the environment terraform code that is executed because having a backend in a module will be ignored when applying.
-

dev.tf > module "Proxy_Instance"

```tf
terraform {
  backend "s3" {
    bucket         = "s3backend-9652"
    key            = "terraform.tfstate"
    region         = "us-east-1"
    dynamodb_table = "terraform-state-lock-9652"
  }
}

module "Proxy_Instance" {
    source = "../Infrastructure"
}
```

- 

main.tf > resource "aws_instance" "my_ec2" > tags

```tf
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "my_ec2" {
  ami           = "ami-0f4b77a23eac2378a"   # Amazon Linux (us-east-1)
  instance_type = "t2.medium"
  key_name      = "Linux-key"

  user_data = <<-EOF
              #!/bin/bash
              dnf install httpd -y
              systemctl start httpd
              systemctl enable httpd
              echo "Hello from Terraform" > /var/www/html/index.html
              EOF

  tags = {
    Name = "Terra-EC2-HTTPD"
  }
}
```

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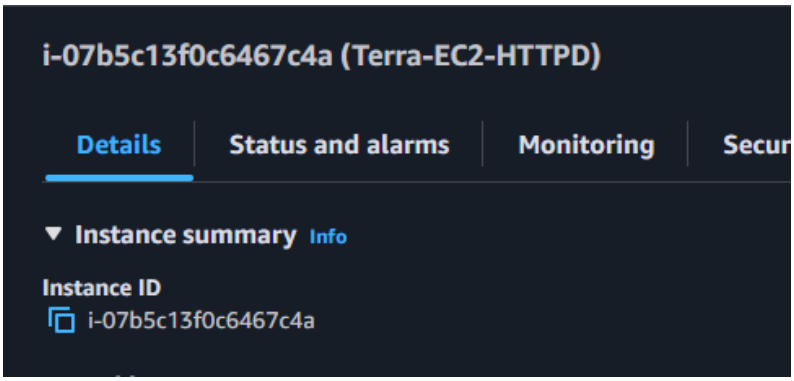
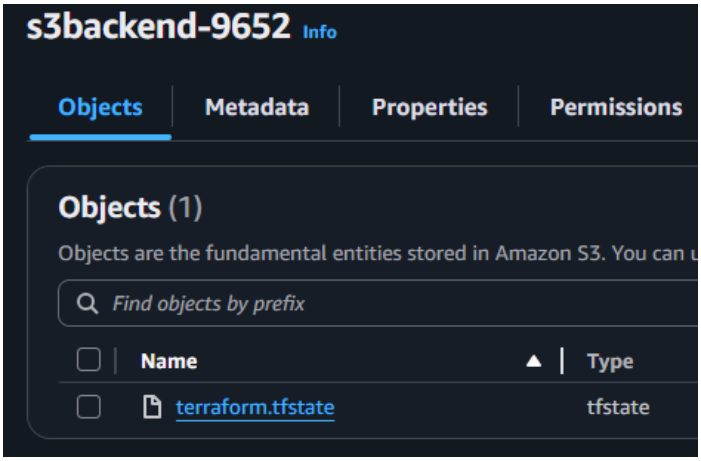
```
    + secondary_network_interface (known after apply)
  }

Plan: 1 to add, 0 to change, 0 to destroy.
module.Proxy_Instance.aws_instance.my_ec2: Creating...
module.Proxy_Instance.aws_instance.my_ec2: Still creating... [00m10s elapsed]
module.Proxy_Instance.aws_instance.my_ec2: Creation complete after 17s [id=i-07b5c13f0c6467c4a]
Releasing state lock. This may take a few moments...

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

-



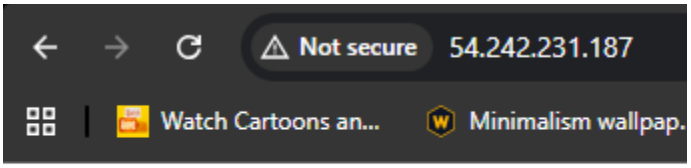
-



-

```
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.s
     Active: active (running) since Thu 2026-03-05 0
       Docs: man:httpd.service(8)
   Main PID: 2761 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 1
      Tasks: 177 (limit: 4666)
```

-



Hello from Terraform

-

- Workspaces:

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- Provision EC2 for 3 different environments (Dev, Staging, and Prod) using Terraform workspaces.

```
PS C:\Users\Viqaas\Desktop\Testing\EC2\Development> terraform workspace new dev
Created and switched to workspace "dev"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Users\Viqaas\Desktop\Testing\EC2\Development> terraform workspace new prod
Created and switched to workspace "prod"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
PS C:\Users\Viqaas\Desktop\Testing\EC2\Development> terraform workspace list
  default
  dev
* prod
```

-
- For the dev workspace:

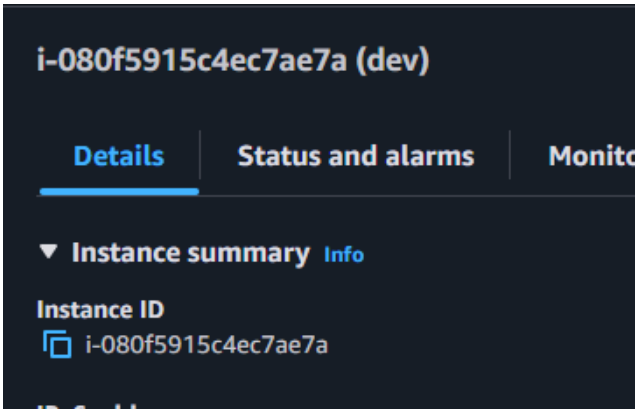
```
dev.tf > module "Proxy_Instance"
terraform {
  backend "s3" {
    bucket         = "s3backend-9652"
    key            = "terraform.tfstate"
    region         = "us-east-1"
    dynamodb_table = "terraform-state-lock-9652"
  }
}

module "Proxy_Instance" {
  source        = "../Infrastructure"
  instance_name = "dev"
}
```

-

```
Plan: 1 to add, 0 to change, 0 to destroy.
module.Proxy_Instance.aws_instance.my_ec2: Creating...
module.Proxy_Instance.aws_instance.my_ec2: Still creating... [00m10s elapsed]
module.Proxy_Instance.aws_instance.my_ec2: Creation complete after 17s [id=i-080f5915c4ec7ae7a]
Releasing state lock. This may take a few moments...
```

-



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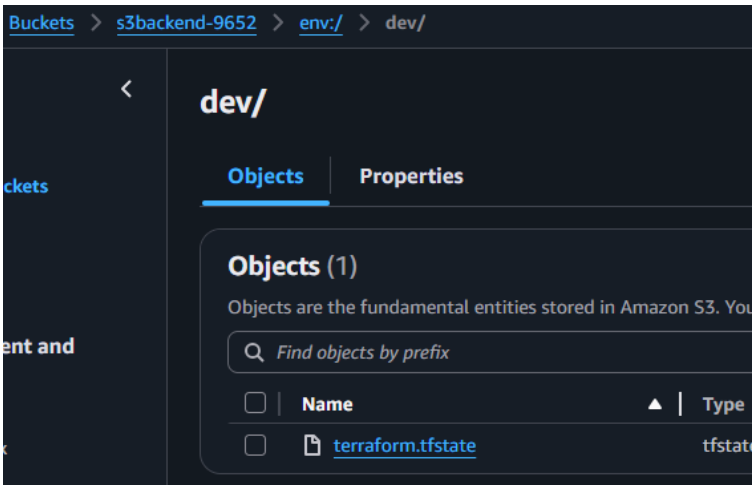
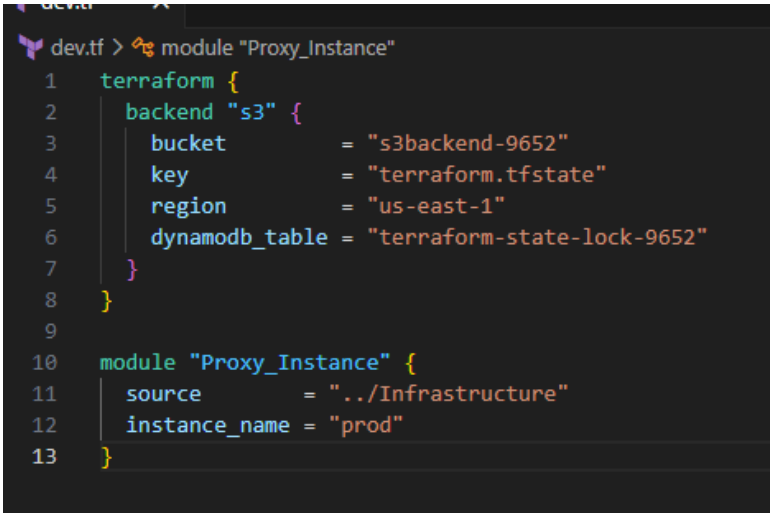
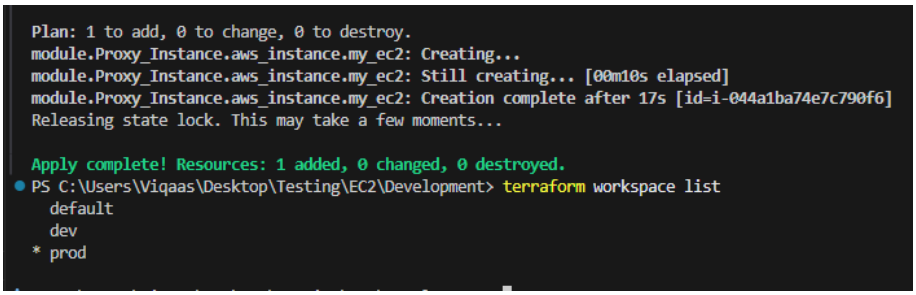
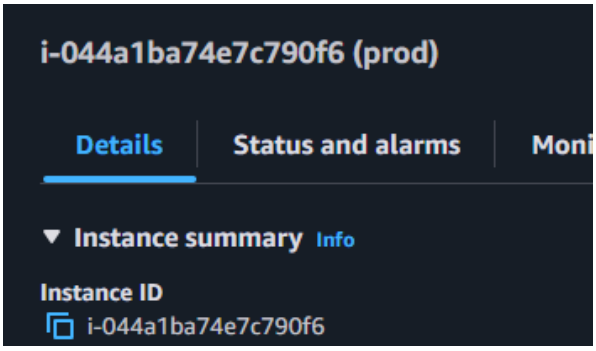
- 

- For the prod workspace:
- 

- 

- 

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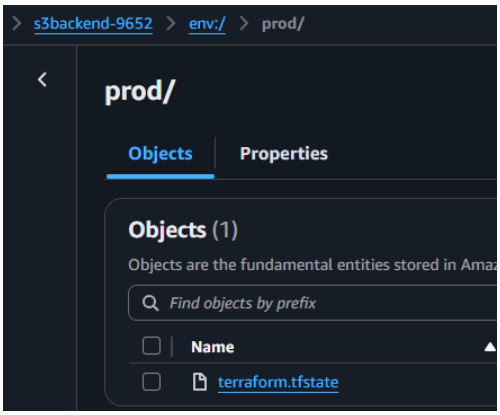
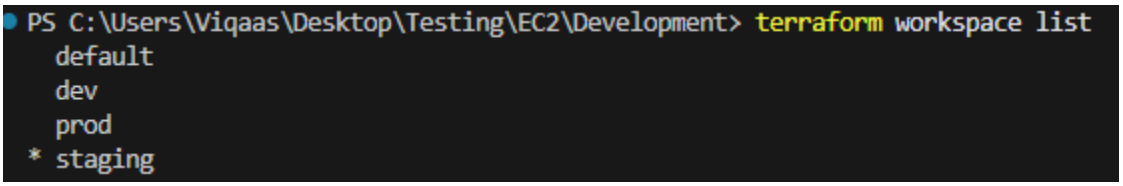
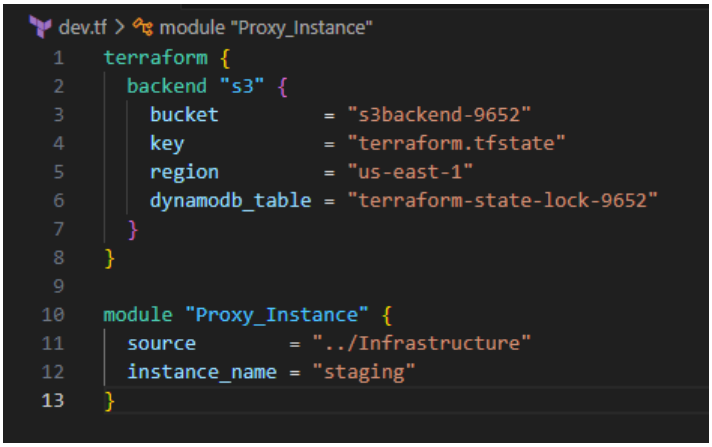
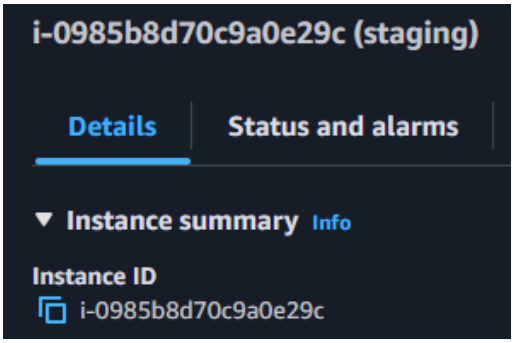
- 

- For the staging workspace:
- 

- 

- 

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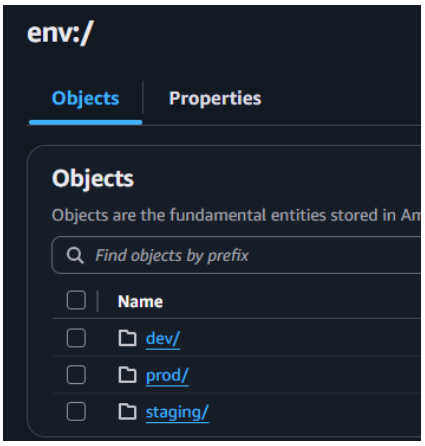
- 

- Provision EC2, S3, and VPC using Terraform modules.
- main.tf > resource "aws_instance" "test"

```
provider "aws" {
    region = "us-east-1"

}
module "vpc" {
    source = "./modules/vpc"
}
module "s3" {
    source = "./modules/s3"
}
resource "aws_instance" "test" {
    ami           = "ami-0f4b77a23eac2378a"
    instance_type = "t2.medium"
    key_name = "Linux-key"
    subnet_id = module.vpc.subnet_id
    user_data = <<-EOF
                #!/bin/bash
                sudo dnf update -y
                sudo dnf install httpd -y
                sudo systemctl start httpd
                EOF

    tags = {
        Name = "TestInstance"
    }

}
```

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- modules > s3 > s3.tf > resource "aws_s3_bucket" "s3_bucket"

```
resource "aws_s3_bucket" "s3_bucket" {
  bucket = "testbench-7869"

  tags = {
    Name = "TestBenchBucket"
  }
}
```

- modules > vpc > vpc.tf > resource "aws_subnet" "mysubnet"

```
resource "aws_vpc" "myvpc" {
  cidr_block = "10.0.0.0/16"

  tags = {
    Name = "MyVPC"
  }
}

resource "aws_subnet" "mysubnet" {
  vpc_id     = aws_vpc.myvpc.id
  cidr_block = "10.0.1.0/24"

  tags = {
    Name = "MySubnet"
  }
}
```

- i-02a285bbab2c7e784 (TestInstance)

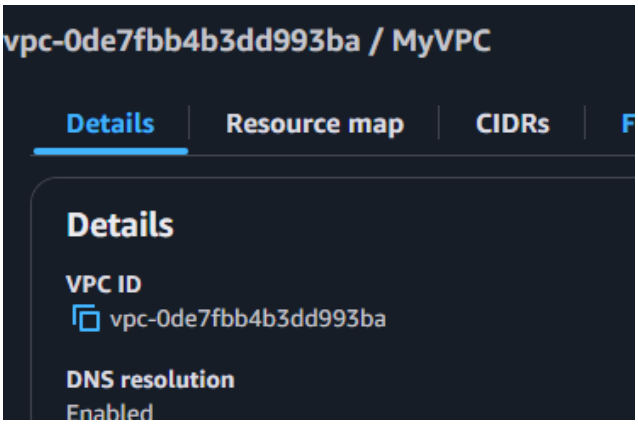
Details | Status and alarms | Monitoring

▼ Instance summary Info

Instance ID
i-02a285bbab2c7e784

IPv6 address

- s3backend-9652
  testbench-7869
- testflow-09

Terraform Task - 06

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